EXECUTIVE SUMMARY

Ma’aden N2 Cluster

New Ammonia Plant Project





As Executive Sponsor, I am pleased to support Ma'aden's vision to advance its ar capabilities and strengthen its position as a global fertilizer leader.

The **N2 Cluster Project** *is a major strategic milestone for Ma'aden and the King Arabia, reinforcing the Kingdom's role in the global fertilizer market and suppor food security.*

KBR is honored to collaborate with Ma'aden and contribute our industry-leadin technology. Backed by decades of proven experience, our solutions deliver relia efficient, and cost-effective performance for world-scale facilities. We are comm being a trusted technology partner, supporting safe, efficient, and successful pr execution.

We will deploy our strongest technical expertise and leverage KBR's experience i ammonia process design and plant optimization. By combining proven technology with best-in-class engineering, we aim to enhance plant reliability, improve energy efficiency, and support Ma'aden's long-term economic and sustainability goals.

We look forward to partnering with Ma'aden and contributing to its continued growth and success.

Hari Ravindran
SVP & Global Head | Technology

Beyond our global project execution record and proven ammonia performance, the Purifier® process offers technical advantages when compared with conventional ammonia technologies that could be of value to Maaden.

Lower Capex

Lower capex in the range of 8-10% than conventional plants due to 35% smaller primary reformer and 12-15% smaller ammonia synloop.

mmonia production

gdom of Saudi
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PURIFIER® TRACK RECORD

GLOBAL MARKET LEADERS

#1 AMMONIA LICENSOR

>96% ON-STREAM FACTOR

350+ DAYS/ YEAR

40+ YEARS OF PURIFIER®

Lowest Energy in Hot Ambient conditions

Cryogenic purification removes inerts before the synthesis loop—an advantage in Ras Al Khair conditions where alternatives face lifecycle efficiency losses. It also delivers the lowest proven energy use, about 5–6% better than conventional ammonia plants.

Demonstrated Reliability at Scale

KBR Purifier plants achieve >96% on-stream time, often exceeding 350 operating days per year. Reliability is enhanced by a mildly fired primary reformer, lower secondary reformer temperatures, and reduced synthesis loop pressure.

Asset Performance Solutions

In-Depth and regional presence to support a 3,500 MTPD mega-train through licensing, commissioning, startup, and the full operating life of the plant.

Ammonia Projects
Track Record

KBR has a proven track record of delivering large and complex grey ammonia projects across the world including KSA. These projects include complex interfaces, local regulatory requirements, protection of licensor technology. Our tools, best practices and systems allow us to transfer knowledge, lessons learnt and best practice as well as de-risk potential threats and realize opportunities to your project.

REFERENCE CASES DIRECTLY RELEVANT TO RAS AL KHAIR

REFERENCE	REGION & CONDITIONS	PERFORMANCE HIGHLIGHT	RELEVANCE TO MA'ADEN
EuroChem	Russia	Operating above design capacity since commissioning in 2019	Industry benchmark for Purifier® performance at scale
OCI Beumont Low Carbon Ammonia	USA	Performance test successfully completed	Proven Synloop operation for 3000 MTPD
Harvest Low Carbon Ammonia	UAE	Under construction	Large Scale Synloop 3000 MTPD
Indorama Train 1, 2, 3	Nigeria	Commissioning of Train 3 completed in 10days from first drop in	Efficient plant with strong economics, client continuously selecting KBR



+260

LICENSED UNITS

Eurochem Ammonia Plant
Performance Highlights

BUILT FOR RELIABILITY

KBR's robust design achieves superior uptime—unlocking higher annual performance from lower capacity and reducing overall CAPEX requirements.

999,364 MT

MAXIMUM ANNUAL PRODUCTION

2021

97.8%

BEST ONSTREAM EFFICIENCY

Jun-Dec 2019

101%

BEST ANNUAL PRODUCTION

2019 & 2021

110%

PEAK LOAD ACHIEVED

In 2021

344 DAYS

MAXIMUM ONSTREAM DAYS

2020

100%

CURRENT OPERATING RATE

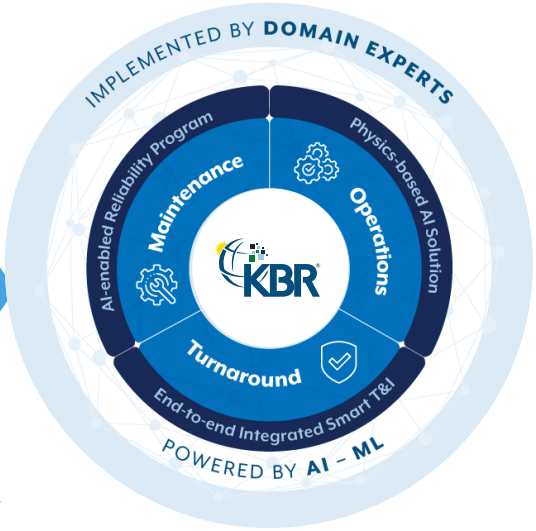
Since Early 2024

- Plant has a nameplate capacity of **2700 MTPD** with design capacity of **2890 MTPD**.
- Performance test for the plant was completed in June 2019 and it clocked and onstream efficiency of **97.8%** till 31st December 2019 with average capacity utilization of **101%**. Highest plant load achieved was **103%**.
- In the year 2020, plant onstream efficiency was **94%** with average annual production of **100%** and highest plant load of **107%** was achieved.
- In the year 2021, plant onstream efficiency was **93%** with average annual production of **101%** and highest plant load of **110%** was achieved.

Driving
Operational Excellence

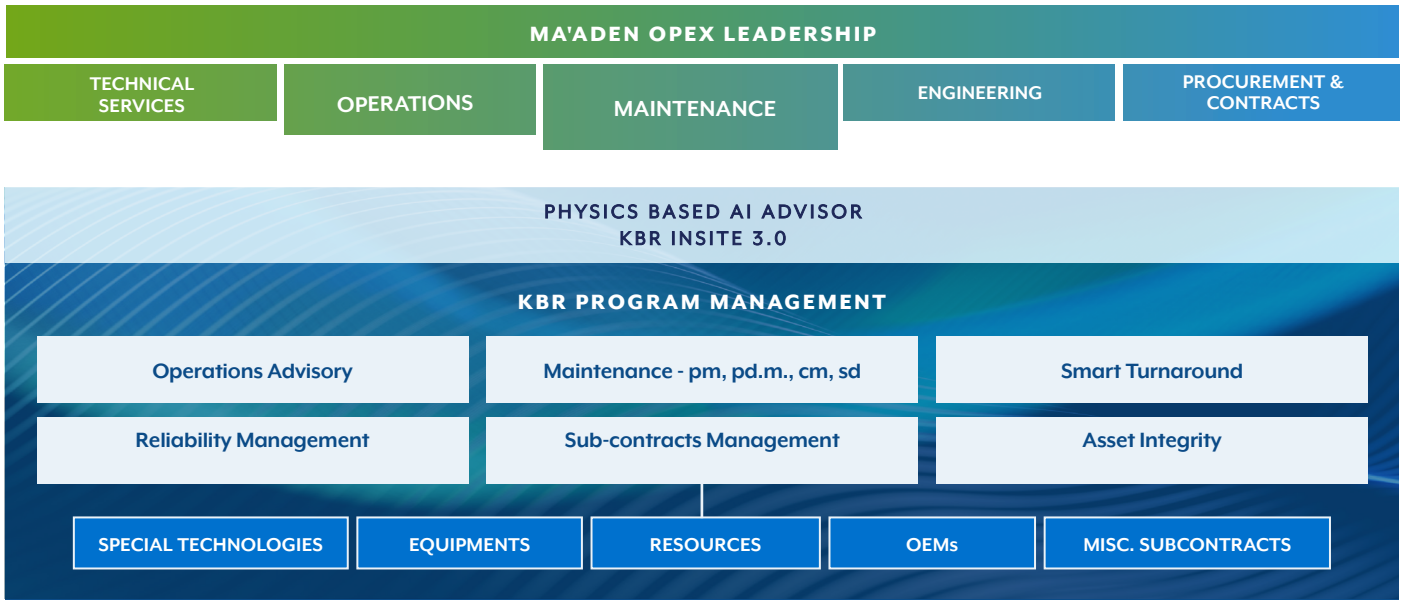
KBR's Operational Excellence Framework





Integrator Business Model

To translate these priorities into sustained operational results, KBR will deploy an Integrator Business Model that unifies execution, reliability, digital tools and specialist capabilities under one performance-led structure.



Next-gen Digital Tools

Real-World Implementation

At the core of this model are three next-generation digital and operational enablers that act as the key differentiators of KBR's Operational Excellence Framework.



INSITE 3.0

- KBR Engineering & Domain Expertise Integrated with advanced AI
- Leverages 100% of your data for decision making
- Predict anomalies early and optimizes in real time
- Provides fully explainable, interactive, physics-based recommendations and advisory



Intelligent Asset Management

- Reduce maintenance budgets and Inventory Management by IAM Analytics
- Optimized Stock by setting Min/Max levels and reorder points

5 Offstream Days, 1% Energy Reduction
1% Yield Improvement through Smart Operational Excellence

20% Inventory Reduction
through optimization and reduced workhours

KBR Benefits

Delivering Measurable Results Across Global Assets

30%
opex savings

Up to **5%**
increase in uptime

20%
inventory reduction

1%
energy reduction

Up to **54%**
Saudization

Trusted Globally, Proven Locally

KBR brings 60+ years of downstream experience, backed by 14,000+ professionals managing critical U.S. assets globally

DUPONT · ALBEMARLE · BASF We create chemistry · SOLVAY · AkzoNobel

ZERO HARM COURAGE TO CARE — Leading Health and Safety Record

satorp — 10 Best Contractor for Safety Awards

Sadara — Four Sadara CEO Eh&S Awards

Shell — We Care/Courage to Care VP Award


SATORP
Long-Term Association


SATORP
Turnaround Award 2025


SADARA
Worlds Largest Petchem


PETRO RABIGH
World's largest Ref - Petchem


DFPCL
INSITE –Digital Smart Factory Award


PTTGC
AI Inventory Optimization


SHELL GTL
Worlds Largest Gas to Liquid


QAFCO
Long-Term GMS Association

Within the Kingdom, our past work includes

Al Jubail ABB Ammonia Plant	1,000 MTPD	1983
Al Jubail IBB Ammonia Plant	1,500 MTPD	1988
Safco 3 Ammonia Plant	1,500 MTPD	2000
IBB Ammonia Plant Revamp	1,500 → 1,650 MTPD	2019
SAFCO-3 Ammonia Plant Revamp	1,500 → 1,800 MTPD	2020

EPC
Partners

KBR has executed **>260 ammonia plants** on all continents and has a joint wealth of experience in engineering and project execution with our EPC partners. The large EPC partner network of KBR opens access to an unmatched number of fabrication yards around the globe, allowing smooth and fast realization of stick-built plant as well as modular solutions for green and blue ammonia.

Why KBR?

Proven World-Scale Technology

Leading ammonia licensor delivering reliable performance across a range of operating conditions.

Energy Efficiency & Lower Operating Costs

Lowest energy consumption reducing fuel usage and emissions while optimizing lifecycle operating costs.

Extensive Experience in the Saudi Arabia

KBR has a track record of successful project delivery in Saudi Arabia, ensuring strong understanding of local execution requirements.

High Reliability & Operational Excellence

Proven designs for maximum uptime, ease of operation, and maintainability, helping plants achieve stable long-term performance.

Trusted Strategic Technology Partner

Continous technical support, digital solutions, and operational guidance—ensuring safe, efficient project execution.

ABOUT KBR, INC.

We deliver science, technology and engineering solutions to governments and companies around the world. KBR employs approximately 36,000 people worldwide with customers in more than 85 countries and operations in over 28 countries. KBR is proud to work with its customers across the globe to provide technology, value-added services, and long-term operations and maintenance services to ensure consistent delivery with predictable results.

At KBR, We Deliver.

kbr.com

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